

# Mecas: A Knowledge Graph Framework for the Startup Ecosystem

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## 1. Graph Design Decisions

Our goal was to create a graph that captures relationships across entities in the startup ecosystem—startups, founders, venture capital firms, industries, and technologies. The graph was built using the `networkx` Python library as a heterogeneous, undirected graph where:

- Nodes represent entities (e.g., a startup, founder, VC, or technology).
- Edges represent meaningful relationships (e.g., a founder founded a startup, a VC invested in a startup, a startup uses a technology).

Each node is typed via attributes (e.g., `type: founder`), allowing us to filter and constrain operations for more targeted similarity or query results. This design offers flexibility to scale or transform the graph into a more complex knowledge graph (e.g., RDF or Neo4j) later.

### Schema Overview

- **Founder** –[founded]– **Startup**
- **Founder** –[has]– **Education, Experience, Background, Expertise**
- **VC** –[invested]– **Startup**
- **VC** –[has]– **Thesis, Geographic Focus**
- **Startup** –[uses]– **Technology**
- **Startup** –[categorized as]– **Industry, Stage, Business Model, Team Size**

### Datasets Used

All data is preprocessed and stored under the `data/` folder with cleaned CSVs:

- `startup_ecosystem_startups.csv`
- `startup_ecosystem_founders.csv`
- `startup_ecosystem_founder_startup.csv`

- `startup_ecosystem_vcs.csv`
- `startup_ecosystem_investments.csv`
- `startup_ecosystem_technologies.csv`
- `startup_ecosystem_startup_tech.csv`

All preprocessing (e.g., exploding list fields, binning numeric values) is handled in `build_graph.py`.

## 2. Similarity Algorithms

We use **SimRank** to calculate entity similarity. SimRank operates on the principle that “two nodes are similar if they are connected to similar nodes.” This is especially powerful in a startup graph, where shared co-founders, investors, or technologies can reveal deeper entity-level similarity.

We additionally support a weighted SimRank mechanism to prevent domination of high-frequency edge types (e.g., excessive investments overpowering thesis or geography for VC similarity).

### Workflow

1. Load the graph with typed relationships using `build_graph.py`.
2. Use `get_similarity_dict(task)` to compute SimRank between entities of interest (e.g., startups).
3. Store and reuse these dictionaries to support fast querying.
4. Use `top_n_similar(entity)` to retrieve the top- $k$  most similar nodes and format for human-readable output.

Each similarity task (e.g., `startup_similarity`, `founder_similarity`) is precomputed and context-aware, leveraging only relevant subgraphs.

## 3. Supported Query Types

The system currently supports the following 8 categories of user queries:

- Q1. Find similar startups:** Based on overlapping founders, investors, industries, and technologies.
- Q2. Find similar VCs:** Based on co-investment networks, shared theses, and investment geographies.
- Q3. Find similar founders:** Based on common startups, education, and domain expertise.

- Q4. Identify co-investors:** Given a VC, list most similar co-investing peers. Two different approaches can be implemented here using this framework. First one is SimRank based approach, which is the one that is submitted. The second one is to find the highest numbers of co-investments for each Venture Capital using the 'coinvestors.json' file.
- Q5. List startup technologies:** Return the tech stack used by a given startup.
- Q6. Industry technology trends:** Identify frequently used technologies in a particular industry.
- Q7. Startup counts by filter:** Count startups in a given industry.
- Q8. VC targeting for founders:** Recommend VCs based on successful investments in similar startups.

## 4. Example Queries and Results

**Query 1: Find 5 startups most similar to Robinson in the AI/ML space**

**Response:**

- 1. Miller-Bass (Similarity score: 0.3197)
- 2. Carter-Jones (Similarity score: 0.2713)
- 3. Burch, Moreno and Thomas (Similarity score: 0.2712)
- 4. DigitalLabs (Similarity score: 0.2709)
- 5. RapidTechnologies (Similarity score: 0.2234)

**Query 2: Find founders with similar backgrounds to Amanda Hernandez**

**Response:**

- 1. Victoria Lee (Similarity score: 0.2120)
- 2. Antonio Smith (Similarity score: 0.1921)
- 3. Gregory Harrell (Similarity score: 0.1921)
- 4. Alexander Raymond (Similarity score: 0.1919)
- 5. Anita Herring (Similarity score: 0.1904)

**Query 3: Which VCs have similar investment patterns to Williams Ventures?**

**Response:**

- 1. Black Accelerator (Similarity score: 0.2314)
- 2. Galvan Partners (Similarity score: 0.2307)
- 3. Lewis Fund (Similarity score: 0.2305)
- 4. Henderson Partners (Similarity score: 0.2304)
- 5. Henry Accelerator (Similarity score: 0.2301)

**Query 4: Which companies have similar technology stacks to Robinson?**

**Response:**

- 1. Miller-Bass (Similarity score: 0.3197)
- 2. Carter-Jones (Similarity score: 0.2713)
- 3. Burch, Moreno and Thomas (Similarity score: 0.2712)
- 4. DigitalLabs (Similarity score: 0.2709)
- 5. RapidTechnologies (Similarity score: 0.2234)

Robinson uses:

- ProAPI, HyperDB, Azure, MaxFlow (x3), UltraDB, NeoFlow, ProFlow, Svelte, React Native, UltraLink, ProSync

**Query 5: Which VCs typically co-invest with Williams Ventures?**

**Response:**

- 1. Secure Partners (Similarity score: 0.0544)
- 2. Meta Partners (Similarity score: 0.0482)
- 3. Castillo Partners (Similarity score: 0.0457)
- 4. Quantum Ventures (Similarity score: 0.0446)
- 5. Washington Fund (Similarity score: 0.0433)

**Query 6: If I'm a founder starting a FinTech company, which VCs should I target based on similar successful investments?**

**Response:**

- 1. Smart Capital (Score: 6.000)
- 2. Prime Ventures (Score: 4.000)
- 3. Meta Capital (Score: 4.000)
- 4. Williams Accelerator (Score: 4.000)
- 5. Johnson Labs (Score: 4.000)

**Query 7: How many startups are in the FinTech industry?**

**Response:**

There are **49** startups in the FinTech industry.

## 5. What I'd Improve With More Time

- **Similarity Scores:** As you might have probably noticed, the similarity scores seem to be low. The reason is because SimRank literally computes similarity between neighbor nodes and not prioritizing or setting different weights to types of nodes. I'd like to focus on this problem in order to have healthier metric.
- **Natural Language Understanding:** Replace the current regex-based query parser with a more robust system using a retrieval-augmented generation (RAG) pipeline.
- **Interactive Visualization:** Integrate Neo4j or D3.js for dynamic, visual exploration of the graph.
- **Model-Based Similarity:** Extend similarity beyond graph topology using embeddings (e.g., Node2Vec, TransE, GNNs).
- **Scalability:** Migrate to a graph database (e.g., Neo4j or TigerGraph) and support distributed computation for larger ecosystems.

## References

- Jeh, G., and Widom, J. (2002). SimRank: A Measure of Structural-Context Similarity. In *Proceedings of the Eighth ACM SIGKDD*.
- NetworkX Documentation: <https://networkx.org>